Daniel Rodríguez Carrión

R&D Engineer | Embedded Linux, Robotics and Cybersecurity

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Professional Profile

R&D engineer specializing in **embedded Linux, robotics, and cybersecurity**. Experience developing and validating embedded platforms, from firmware and hardware integration to real-time Linux configuration, ROS 2 integration, and automated testing. Background in advanced sensing, IoT, and security evaluation of connected devices, with a strong focus on reproducible development workflows, requirements traceability, and evidence-based validation.

Professional Experience

Embedded Systems and R&D Engineer FUNDITEC
22/09/2026 – Present
PRIMUS OS Project

- Developed the embedded Linux platform for **PRIMUS OS** on **NVIDIA Jetson Orin NX**, configuring and integrating a **PREEMPT_RT** kernel, a minimal root filesystem, and board support components; verified deployment and boot on target hardware.
- Built a reproducible **Docker-based build and flashing pipeline** using **Bash and Python**, automating kernel configuration checks, BSP integration, QSPI/NVMe provisioning, post-flash verification, and diagnostic evidence collection.
- Implemented a proof of concept combining a process orchestrator, **ROS 2 Humble** nodes, and a mock control interface, exercising the same application bundle in **software-in-the-loop (SIL)** and on the Jetson platform.
- Developed automated integration and fault-injection tests covering nominal execution, process crashes, timeouts, and interface failures, with structured logging and execution identifiers for end-to-end traceability.
- Designed and implemented a **host–Jetson communications benchmark** for ROS 2 and **MuJoCo**-based simulation, covering latency, jitter, message delivery, resource usage, and recovery under degraded communication conditions.
- Produced architecture decision records, requirements and test traceability matrices, validation procedures, and reproducible evidence packages to support the transition from proof of concept to a maintainable embedded platform.

Electronics and R&D Engineer Quandum Aerospace
06/2026 – 09/2026

- First engineer to join the electronics department, contributing to the technical definition and preliminary design of a mobile airport platform aimed at pilot assistance, obstacle detection, and relative positioning.
- Researched, analyzed, and selected sensors suitable for the project, evaluating different perception and detection technologies with consideration of aeronautical standards such as **DO-178** and **DO-278**.
- Started firmware development on the **NXP S32K344** microcontroller, focused on the embedded control architecture and the integration of system peripherals and communications.
- Evaluated an **i.MX8**-based ECU as a high-level processing unit, carrying out tests on the **Apalis Evaluation Board** with **Torizon OS**.

Cybersecurity and R&D Engineer DEKRA
01/2025 – 02/2026

- Participated in the evaluation team for the **SESIP Level 5** certification of **Google pKVM for Android**, helping accelerate testing and delivery cycles through test automation.
- Managed **4 evaluation projects** end to end on embedded and IoT platforms, all successfully completed, covering devices such as **nRF52833**, **RTL907XD**, **ST87M01**, and **Arm MPS3** with **RealSecuriTEE**.
- Assessed the available network and communication interfaces of each device according to project scope, together with validation of critical mechanisms such as **secure boot**, **secure firmware upgrade**, **secure debug**, and **secure storage**.
- Developed tooling in **Python** and **C/C++**, including test automation using the **PSA API**, reducing delivery times for several test campaigns by **1–2 weeks**.
- Produced technical documentation and evidence packages for certification and regulatory assessment processes involving **SESIP** and **RED**.

Mechatronics and Aerial Robotics Engineer AVERON
01/2025 – 05/2025
Project-based collaboration alongside DEKRA

- Provided technical support to an aerial robotics startup, alongside my main role in cybersecurity and R&D.
- Performed conceptual design and 3D modeling of components for UAV platforms using **CATIA V5**.
- Supported mechatronic integration and functional prototyping tasks for autonomous drone solutions.

Technical Staff and Researcher University of Málaga
02/2022 – 09/2024
HortSost Project: project page

- Contributed to technical development in the **HortSost** project, working on sensing, firmware, electronics, and experimental validation of low-power IoT platforms, establishing the technical foundation for my later specialization in embedded assessment and security.
- Deployed sensing solutions for environmental and energy monitoring, including **current, humidity, temperature, anemometry, and rainfall** sensors, together with **electronic signal conditioning** and **PCB design** tailored to each scenario.
- Designed and implemented a self-configuring wireless sensor network infrastructure for **edge computing**, based on **ESP32-C3**, **FreeRTOS**, **ESP-NOW**, and **MQTT**, with a coordinator node, auto-pairing, and Internet gateway capabilities.
- Experimentally evaluated the platform, achieving a **12.5%** reduction in compiled binary size with an optimized **ESP-IDF** configuration, reducing the memory footprint and leaving more room for additional functionality without penalizing execution times.
- Developed software for the **Acconeer A121** radar, focused on biometric readings such as **heart rate** and **respiration** for room climate regulation systems.
- Contributed to research results that led to a **paper and conference presentation** at the **Spanish Congress on Computer Science (CEDI 2024)**.

Technologies

Firmware and embedded	C, C++, ESP32, ESP8266, NXP S32K344, FreeRTOS, OTA, RISC-V
Embedded Linux	NVIDIA Jetson Orin NX, Jetson Linux/L4T, PREEMPT_RT, Kconfig, BSP integration, rootfs customization, systemd, NXP i.MX8, Torizon OS
Robotics and simulation	ROS 2 Humble, MuJoCo, software-in-the-loop (SIL), host–target integration
Automation and testing	Python, Bash, CMake, Makefiles, PSA API, test automation, fault injection, communications benchmarking
IoT and connectivity	MQTT, Node-RED, NB-IoT, BLE, Bluetooth Mesh, Thread, Zigbee, NFC
Networks and buses	Ethernet, MACsec, I2C, JTAG, PCIe
IoT protocols and services	HTTP/HTTPS, CoAP, LwM2M, TCP/IP, TLS/DTLS
Systems and infrastructure	Linux/Ubuntu, Docker, Git, virtual machines, Raspberry Pi
Embedded security	Secure boot, secure storage, secure firmware upgrade, secure debug, applied cryptography
Engineering and validation	Requirements traceability, architecture decision records, structured logging, technical documentation, hardware validation
Design and prototyping	CATIA V5, 3D modeling, mechatronic integration

Education

- **MSc in Mechatronics Engineering** — University of Málaga, 2023
- **BSc in Electronics, Robotics and Mechatronics Engineering** — University of Seville / University of Málaga, 2021
- **Project Management for Photovoltaic Solar Installations** — Euroinnova, 2023

Certifications and Languages

- **CATIA V5 – Part Design Associate** — Dassault Systèmes, 2024
- **RPAS Pilot and Radio Operator** — Aerocámaras, 2019
- **Languages:** Native Spanish, English B2 (Cambridge FCE)

Publication

Self-configuring wireless sensor network for edge computing, Spanish Congress on Computer Science (CEDI), A Coruña, 2024.